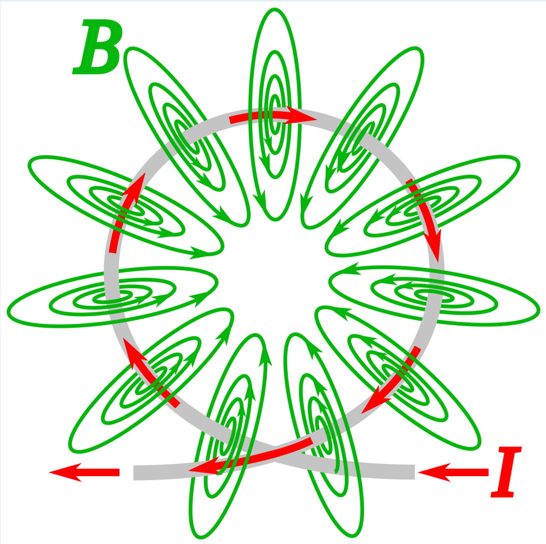
CBSE Class 12 Physics

Chapter 4: Moving Charges and Magnetism

Complete Board Exam Notes with LaTeX Equations and Credited Reference Diagrams

Designed for: CBSE board examination preparation, NCERT-based conceptual understanding, derivations, formula recall, numerical practice, and quick revision.

Includes: overview, derivations, formulas, examples, NCERT-style questions, board-style questions, common mistakes, credited diagrams, and final checklist.



Magnetic field lines around a current loop

Aligned to NCERT Class XII Physics Chapter 4 and CBSE Physics 042 syllabus scope.
Compiled as a polished LaTeX PDF with clean equations and credited reference diagrams.

Contents

Syllabus Alignment and Exam Target

Must Know

This chapter belongs to **Unit III: Magnetic Effects of Current and Magnetism**. CBSE lists Moving Charges and Magnetism with: magnetic field, Oersted's experiment, Biot-Savart law and circular loop, Ampere's law and long straight wire, straight solenoid qualitative treatment, force on moving charge in electric and magnetic fields, force on current-carrying conductor, force between two parallel conductors and definition of ampere, torque on current loop, magnetic dipole moment, moving coil galvanometer, current sensitivity, and conversion to ammeter and voltmeter.

How to Use These Notes

1. First learn the **direction rules**: right-hand rule, Fleming's left-hand rule, dot-cross convention.
2. Memorise the **formula sheet**; it is the fastest way to solve board numericals.
3. Practise all derivations with diagrams. In CBSE answers, a correct labeled diagram often earns marks.
4. Use the final checklist one day before the exam.

Exam focus	Board answer requirement
Definitions	State formula, SI unit, direction rule, and conditions.
Derivations	Draw diagram, mention assumptions, write vector/scalar relation, simplify stepwise.
Numericals	SI conversion, formula substitution, units in final answer.
Reasoning questions	Explain using force direction, symmetry, or field-line logic.

1 Chapter Overview

1.1 Core Idea

Moving charges produce magnetic fields. Magnetic fields exert forces on moving charges and current-carrying conductors. The chapter connects three ideas:

Moving charges / currents	⇒	**Magnetic field B**	⇒	**Force / torque / motion**
Current elements and loops	Biot–Savart law; Ampere's law	Field around wires, loops and solenoids	Lorentz force; conductor force	Circular motion, cyclotron, loop torque, galvanometer

1.2 Main Physical Quantities

Quantity	**Symbol**	**SI Unit**	**Meaning**

Magnetic field	$\boldsymbol{B}$	tesla (T)	Field that exerts force on moving charge/current.
Magnetic force	$\boldsymbol{F}_B$	newton (N)	Force due to magnetic field.
Current element	$I\,\mathrm{d}\boldsymbol{l}$	$\mathrm{A\,m}$	Small element of current-carrying wire.
Magnetic moment	$\boldsymbol{m}$	$\mathrm{A\,m^2}$	Strength and orientation of current loop as dipole.
Permeability of free space	μ_0	$\mathrm{T\,m\,A^{-1}}$	$4\pi \times 10^{-7}\,\mathrm{T\,m\,A^{-1}}$.
Cyclotron frequency	f	hertz (Hz)	Frequency of circular motion in uniform $\boldsymbol{B}$.

1.3 Dot-Cross Convention

Symbol	Meaning
$\odot$	Vector coming **out of the page**; like seeing the arrow tip.
$\otimes$	Vector going **into the page**; like seeing the arrow tail.

Figure 1: Dot-cross convention used throughout magnetism.

Exam Tip

In board diagrams, draw several dots/crosses to show a uniform magnetic field. Label them clearly as $\boldsymbol{B}$ out of page or into page.

2 Magnetic Force on a Moving Charge

2.1 Vector Formula

A charge q moving with velocity $\boldsymbol{v}$ in magnetic field $\boldsymbol{B}$ experiences magnetic force

$$\boxed{\boldsymbol{F}_B = q\,(\boldsymbol{v} \times \boldsymbol{B})} \tag{1}$$

Magnitude:

$$\boxed{F_B = |q|vB\sin\theta} \tag{2}$$

where θ is the angle between $\boldsymbol{v}$ and $\boldsymbol{B}$.

Formula Box

$$\begin{aligned} F_B &= |q|vB\sin\theta, \\ F_B &= 0 \quad \text{if } \theta = 0^\circ \text{ or } 180^\circ, \\ F_B &= |q|vB \quad \text{if } \theta = 90^\circ. \end{aligned}$$

2.2 Direction of Force

For a **positive charge**, use the right-hand rule for $\boldsymbol{v} \times \boldsymbol{B}$. For a **negative charge**, reverse the direction.

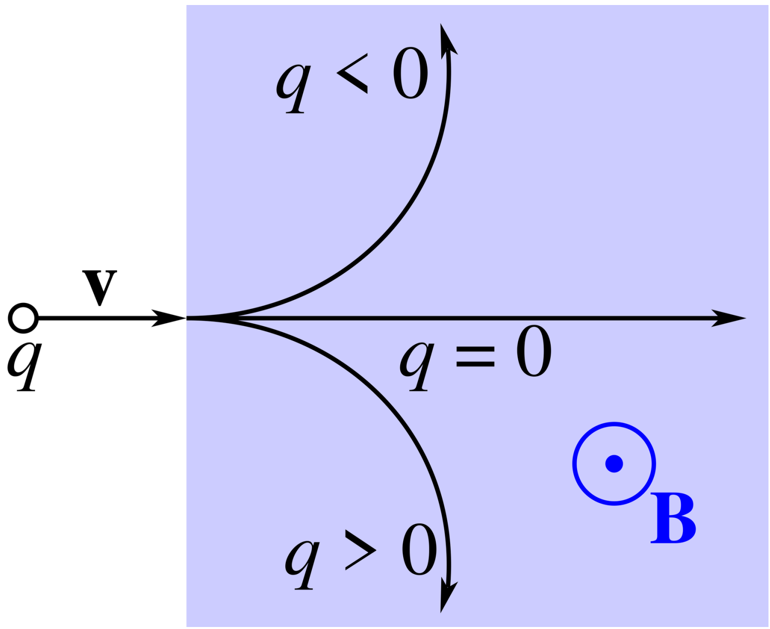


Figure 2: Magnetic force direction for moving charged particles in a uniform magnetic field. Use $\boldsymbol{F}_B = q(\boldsymbol{v} \times \boldsymbol{B})$ and reverse the direction for a negative charge.

2.3 Properties of Magnetic Force

1. Magnetic force is always perpendicular to $\boldsymbol{v}$ and $\boldsymbol{B}$.
2. Magnetic force does no work because $\boldsymbol{F}_B \perp \boldsymbol{v}$.
3. Speed and kinetic energy remain constant in a magnetic field alone.
4. Magnetic field changes only the direction of velocity, not its magnitude.

Why Magnetic Force Does No Work

Work done in time $\mathrm{d}t$ is

$$\begin{aligned} \mathrm{d}W &= \boldsymbol{F}_B \cdot \mathrm{d}\boldsymbol{r} && (3) \\ &= q(\boldsymbol{v} \times \boldsymbol{B}) \cdot (\boldsymbol{v}\,\mathrm{d}t) && (4) \\ &= q\,\mathrm{d}t\,[(\boldsymbol{v} \times \boldsymbol{B}) \cdot \boldsymbol{v}] = 0. && (5) \end{aligned}$$

The scalar product is zero because $\boldsymbol{v} \times \boldsymbol{B}$ is perpendicular to $\boldsymbol{v}$.

3 Lorentz Force

When both electric field $\boldsymbol{E}$ and magnetic field $\boldsymbol{B}$ are present, total force is

$$\boxed{\boldsymbol{F} = q\,(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B})} \tag{6}$$

This is called the **Lorentz force**.

Force	Formula	Important property
Electric force	$\boldsymbol{F}_E = q\boldsymbol{E}$	Can change speed and kinetic energy.
Magnetic force	$\boldsymbol{F}_B = q(\boldsymbol{v} \times \boldsymbol{B})$	Changes direction of motion; no work done.
Lorentz force	$q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B})$	Total force in crossed fields.

3.1 Velocity Selector Condition

In crossed electric and magnetic fields, an undeflected charged particle satisfies

$$qE = qvB \quad \Rightarrow \quad \boxed{v = \frac{E}{B}}. \tag{7}$$

4 Motion of a Charged Particle in a Magnetic Field

4.1 Case 1: Velocity Perpendicular to Magnetic Field

If $\boldsymbol{v} \perp \boldsymbol{B}$, the magnetic force provides centripetal force:

$$|q|vB = \frac{mv^2}{r} \tag{8}$$

$$r = \boxed{\frac{mv}{|q|B}} \tag{9}$$

Angular speed:

$$\boxed{\omega = \frac{|q|B}{m}} \tag{10}$$

Time period:

$$\boxed{T = \frac{2\pi m}{|q|B}} \tag{11}$$

Frequency:

$$\boxed{f = \frac{|q|B}{2\pi m}} \tag{12}$$

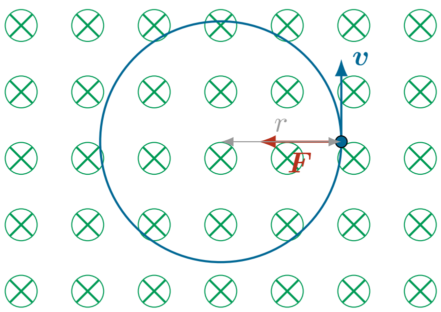


Figure 3: Magnetic force acts as centripetal force. It is always perpendicular to velocity.

4.2 Case 2: Velocity Parallel to Magnetic Field

If $\boldsymbol{v} \parallel \boldsymbol{B}$, then $\theta = 0^\circ$ and

$$F_B = |q|vB \sin 0^\circ = 0. \tag{13}$$

The particle moves in a straight line with constant velocity.

4.3 Case 3: Oblique Velocity - Helical Motion

If velocity has two components,

$$\boldsymbol{v} = \boldsymbol{v}_\perp + \boldsymbol{v}_\parallel, \tag{14}$$

then $v_\perp$ causes circular motion and $v_\parallel$ produces uniform motion along the field. The path is a helix.

$$r = \frac{mv_\perp}{|q|B}, \tag{15}$$

$$T = \frac{2\pi m}{|q|B}, \tag{16}$$

$$\text{pitch } p = v_\parallel T = \frac{2\pi m v_\parallel}{|q|B}. \tag{17}$$

Helical path when $\boldsymbol{v}$ has perpendicular and parallel components

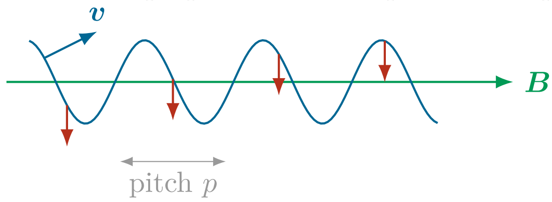


Figure 4: Helical motion: circular motion around the field direction plus forward drift.

Exam Tip

The cyclotron frequency is independent of speed and radius only in the non-relativistic limit. This is often asked as a conceptual reason behind cyclotron resonance.

5 Cyclotron

5.1 Principle

A cyclotron accelerates charged particles using a high-frequency alternating electric field and a perpendicular magnetic field. The magnetic field bends the particle into circular arcs; the electric field accelerates it whenever it crosses the gap between the two dees.

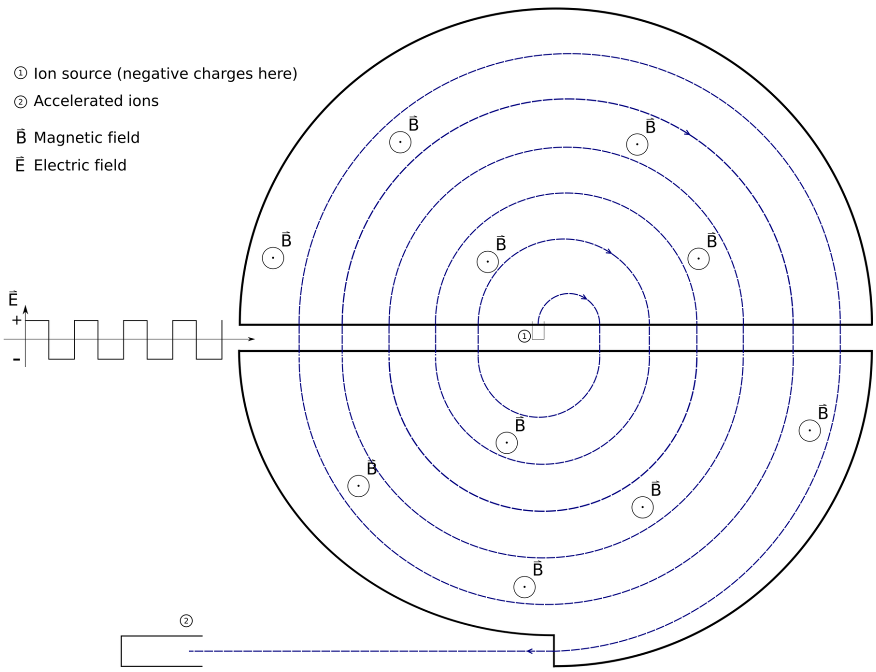


Figure 5: Cyclotron reference diagram: dees, alternating electric field across the gap, perpendicular magnetic field, and spiral particle path.

5.2 Cyclotron Frequency

Inside a dee, magnetic force provides centripetal force:

$$|q|vB = \frac{mv^2}{r}, \tag{18}$$

$$\omega = \frac{v}{r} = \frac{|q|B}{m}. \tag{19}$$

Therefore,

$$\boxed{f = \frac{\omega}{2\pi} = \frac{|q|B}{2\pi m}}. \tag{20}$$

5.3 Maximum Energy

If R is the maximum radius of the dee,

$$v_{\max} = \frac{|q|BR}{m}, \tag{21}$$

$$K_{\max} = \frac{1}{2}mv_{\max}^2 = \boxed{\frac{q^2B^2R^2}{2m}}. \tag{22}$$

5.4 Limitations

- It cannot accelerate neutral particles.
- It is unsuitable for electrons at high speeds because relativistic mass variation changes the frequency.
- The frequency condition must match the cyclotron frequency.

6 Biot-Savart Law

6.1 Statement

The magnetic field $\mathrm{d}\boldsymbol{B}$ at a point due to a small current element $I\,\mathrm{d}\boldsymbol{l}$ is

$$\boxed{\mathrm{d}\boldsymbol{B} = \frac{\mu_0}{4\pi}\frac{I\,\mathrm{d}\boldsymbol{l} \times \hat{\boldsymbol{r}}}{r^2}} \tag{23}$$

Magnitude:

$$\boxed{\mathrm{d}B = \frac{\mu_0}{4\pi}\frac{I\,\mathrm{d}l\sin\theta}{r^2}} \tag{24}$$

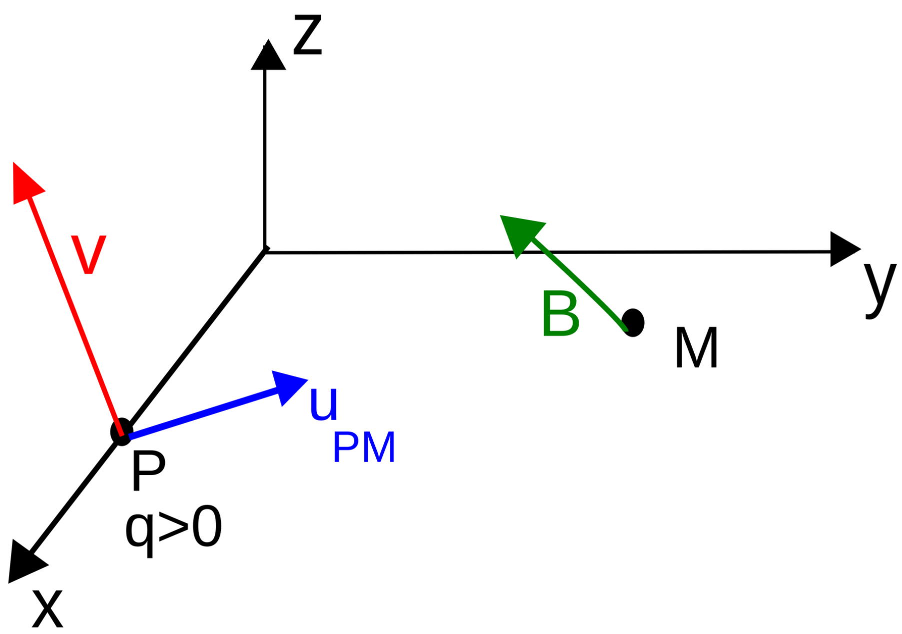


Figure 6: Geometry of the Biot–Savart law for a current element and observation point.

6.2 Comparison with Coulomb's Law

	Coulomb's law	**Biot-Savart law**
Source	Charge q	Current element $I\,\mathrm{d}\boldsymbol{l}$
Field	Electric field $\boldsymbol{E}$	Magnetic field $\boldsymbol{B}$
Dependence	$E \propto 1/r^2$	$\mathrm{d}B \propto 1/r^2$
Direction	Along $\boldsymbol{r}$	Perpendicular to plane of $\mathrm{d}\boldsymbol{l}$ and $\boldsymbol{r}$

7 Magnetic Field on the Axis of a Circular Current Loop

7.1 Result

For a circular loop of radius R, carrying current I, magnetic field at a point on the axis at distance x from the centre is

$$\boxed{B = \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}}} \tag{25}$$

At the centre, $x = 0$:

$$\boxed{B_{\text{centre}} = \frac{\mu_0 I}{2R}}. \tag{26}$$

For N turns:

$$\boxed{B = \frac{\mu_0 N I R^2}{2(R^2 + x^2)^{3/2}}}. \tag{27}$$

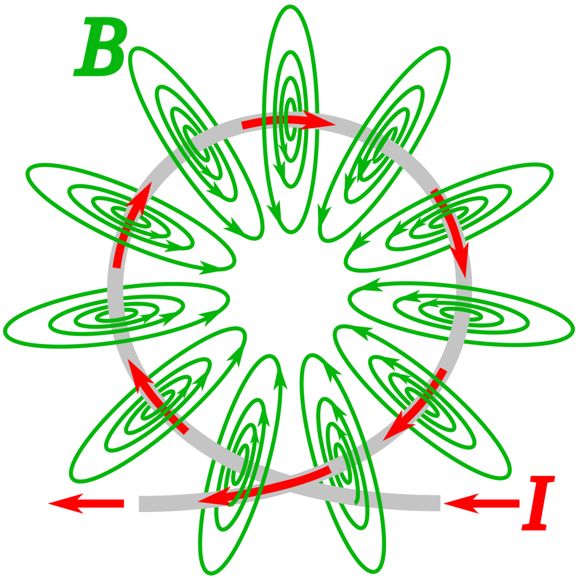


Figure 7: Magnetic field pattern of a circular current loop. On the axis, transverse components cancel by symmetry and axial components add.

Derivation: Field on Axis of Circular Loop

For a current element $I\,\mathrm{d}l$ on the loop, Biot-Savart law gives

$$\mathrm{d}B = \frac{\mu_0}{4\pi}\frac{I\,\mathrm{d}l\sin 90^\circ}{r^2} = \frac{\mu_0}{4\pi}\frac{I\,\mathrm{d}l}{r^2}. \tag{28}$$

Here

$$r = \sqrt{R^2 + x^2}. \tag{29}$$

Only the axial component survives due to symmetry:

$$\mathrm{d}B_{\text{axis}} = \mathrm{d}B\cos\alpha. \tag{30}$$

From the geometry,

$$\cos\alpha = \frac{R}{r} = \frac{R}{\sqrt{R^2 + x^2}}. \tag{31}$$

Therefore,

$$B = \int \mathrm{d}B\cos\alpha \tag{32}$$

$$= \int \frac{\mu_0}{4\pi}\frac{I\,\mathrm{d}l}{r^2}\frac{R}{r} \tag{33}$$

$$= \frac{\mu_0 IR}{4\pi r^3}\int \mathrm{d}l \tag{34}$$

$$= \frac{\mu_0 IR}{4\pi(R^2 + x^2)^{3/2}}(2\pi R) \tag{35}$$

$$= \boxed{\frac{\mu_0 IR^2}{2(R^2 + x^2)^{3/2}}}. \tag{36}$$

8 Ampere's Circuital Law

8.1 Statement

Ampere's circuital law states that the line integral of magnetic field around any closed path is μ_0 times the net current passing through the area bounded by the path:

$$\boxed{\oint \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{l} = \mu_0 I_{\text{enclosed}}}. \tag{37}$$

Must Know

Use Ampere's law when symmetry makes B constant along a convenient closed path: infinitely long straight wire, ideal solenoid, toroid.

9 Magnetic Field Due to a Straight Current-Carrying Wire

9.1 Using Ampere's Law: Infinite Straight Wire

For an infinitely long straight conductor carrying current I, take a circular Amperian loop of radius r around the wire. By symmetry, B is constant and tangential.

$$\oint \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{l} = B \oint \mathrm{d}l = B(2\pi r), \tag{38}$$

$$B(2\pi r) = \mu_0 I, \tag{39}$$

$$B = \boxed{\frac{\mu_0 I}{2\pi r}}. \tag{40}$$

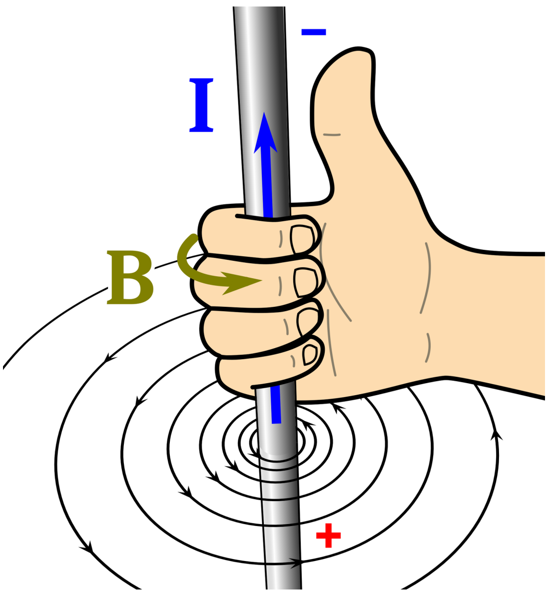


Figure 8: Right-hand grip rule for magnetic field lines around a long straight current-carrying wire.

9.2 Direction Rule

Grip the wire with your right hand, thumb along current direction. Curled fingers give the direction of magnetic field lines.

9.3 Finite Straight Wire Result

For a finite wire, field at perpendicular distance r is

$$\boxed{B = \frac{\mu_0 I}{4\pi r}(\sin\theta_1 + \sin\theta_2)}. \tag{41}$$

For an infinitely long wire, $\theta_1 = \theta_2 = 90^\circ$ and the result becomes $B = \mu_0 I/(2\pi r)$.

10 Straight Solenoid: Qualitative Treatment

A solenoid is a long coil of many closely wound turns. Inside a long solenoid, magnetic field is nearly uniform and parallel to the axis. Outside, the field is weak for an ideal long solenoid.

$$\boxed{B \approx \mu_0 n I \quad \text{inside a long air-core solenoid}} \tag{42}$$

where $n = N/L$ is number of turns per unit length.

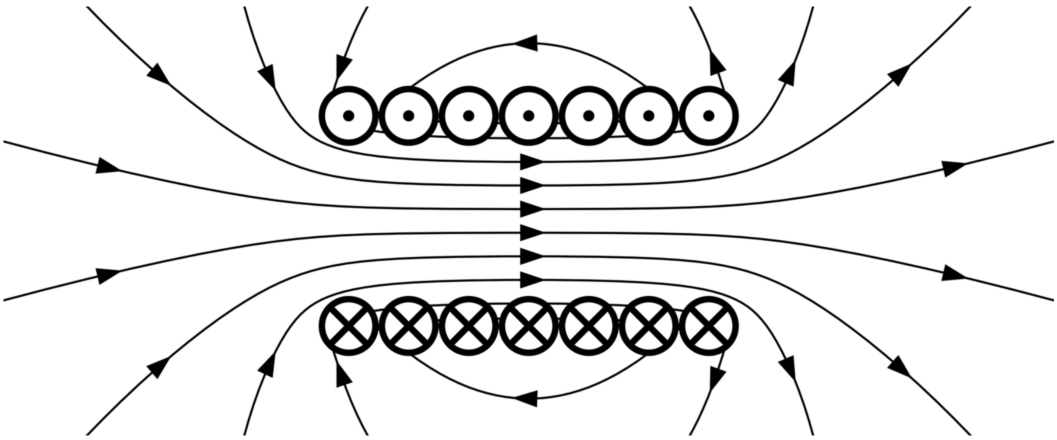


Figure 9: Magnetic field pattern of a long solenoid. Inside a long solenoid, field lines are nearly parallel and uniform.

11 Force on a Current-Carrying Conductor

A current-carrying conductor in a uniform magnetic field experiences force because charges inside it are moving.

$$\boxed{\boldsymbol{F} = I(\boldsymbol{L} \times \boldsymbol{B})} \tag{43}$$

Magnitude:

$$\boxed{F = ILB\sin\theta} \tag{44}$$

where $\boldsymbol{L}$ is length vector in the direction of current.

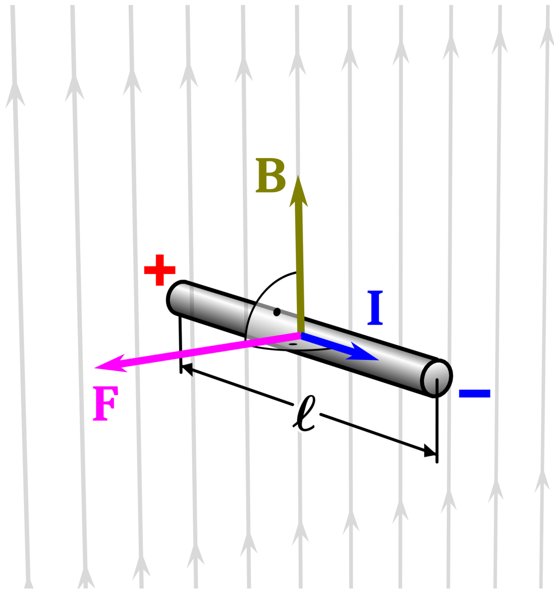


Figure 10: Force on a current-carrying conductor in a magnetic field: $\boldsymbol{F} = I(\boldsymbol{L} \times \boldsymbol{B})$.

12 Force Between Two Parallel Current-Carrying Wires

12.1 Derivation

Consider two long parallel conductors separated by distance d, carrying currents I_1 and I_2. Magnetic field produced by wire 1 at the position of wire 2 is

$$B_1 = \frac{\mu_0 I_1}{2\pi d}. \tag{45}$$

Force on length L of wire 2:

$$F_2 = I_2 L B_1. \tag{46}$$

Therefore,

$$\boxed{\frac{F}{L} = \frac{\mu_0 I_1 I_2}{2\pi d}}. \tag{47}$$

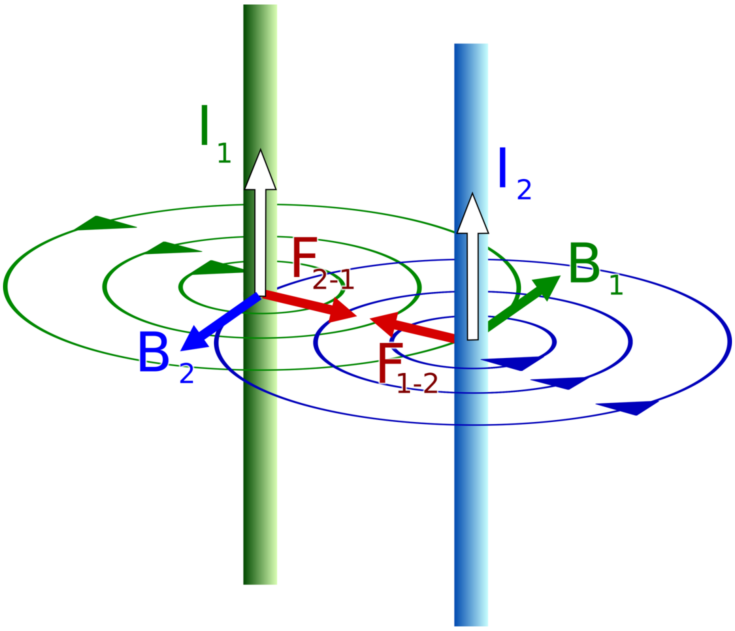


Figure 11: Attraction between two long parallel wires carrying currents in the same direction. If one current is reversed, the force becomes repulsive.

12.2 Definition of Ampere

One ampere is the current which, when maintained in each of two long, straight, parallel conductors of negligible cross-section placed one metre apart in vacuum, produces a force of $2 \times 10^{-7}\,\mathrm{N\,m^{-1}}$ between them.

13 Torque on a Current Loop

13.1 Magnetic Dipole Moment

For a current loop,

$$\boxed{\boldsymbol{m} = NI\boldsymbol{A}} \tag{48}$$

where $\boldsymbol{A}$ is area vector normal to the plane of the loop. Magnitude:

$$\boxed{m = NIA}. \tag{49}$$

13.2 Torque Formula

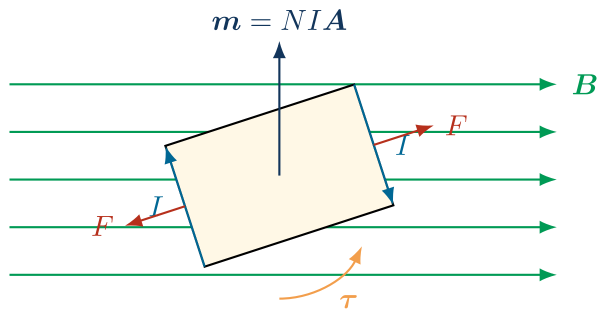
A current loop in a uniform magnetic field experiences torque:

$$\boxed{\boldsymbol{\tau} = \boldsymbol{m} \times \boldsymbol{B}} \tag{50}$$

Magnitude:

$$\boxed{\tau = NIAB\sin\theta} \tag{51}$$

where θ is angle between $\boldsymbol{m}$ and $\boldsymbol{B}$.



Equal and opposite magnetic forces form a couple.

Figure 12: A rectangular current loop in a uniform magnetic field experiences a torque.

Derivation: Torque on Rectangular Current Loop

Let a rectangular coil have N turns, sides a and b, current I, and area $A = ab$. In a uniform magnetic field B, forces on two opposite sides form a couple.
Force on one side of length a:

$$F = IaB. \tag{52}$$

The perpendicular separation between the two forces is $b \sin\theta$. Hence torque for one turn:

$$\begin{aligned} \tau &= F(b\sin\theta) &(53)\\ &= IaB(b\sin\theta) &(54)\\ &= IAB\sin\theta. &(55)\end{aligned}$$

For N turns,

$$\boxed{\tau = NIAB\sin\theta}. \tag{56}$$

Since $m = NIA$,

$$\boxed{\tau = mB\sin\theta}. \tag{57}$$

14 Moving Coil Galvanometer

14.1 Purpose and Principle

A moving coil galvanometer detects small currents. It works on the principle that a current-carrying coil placed in a magnetic field experiences torque.

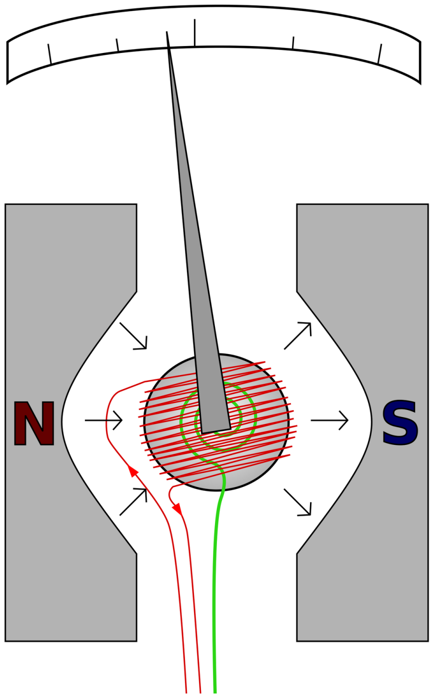


Figure 13: Moving coil galvanometer reference diagram showing coil, pointer, spring, and magnetic pole pieces.

14.2 Theory

Deflecting torque on the coil is

$$\tau_d = NIAB. \tag{58}$$

Restoring torque due to suspension fibre is

$$\tau_r = k\phi, \tag{59}$$

where ϕ is angular deflection and k is torsional constant. At equilibrium,

$$NIAB = k\phi, \tag{60}$$

$$I = \boxed{\frac{k}{NAB}\phi}. \tag{61}$$

Thus, current is directly proportional to deflection:

$$\boxed{I \propto \phi}. \tag{62}$$

14.3 Current Sensitivity

Current sensitivity is deflection per unit current:

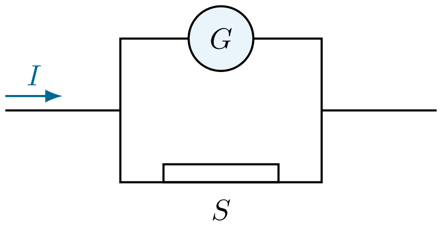
$$\boxed{S_I = \frac{\phi}{I} = \frac{NAB}{k}}. \tag{63}$$

It increases when N, A, or B increases, or k decreases.

14.4 Conversion to Ammeter

A galvanometer of resistance G and full-scale current I_g is converted to an ammeter of range I by connecting a low shunt resistance S in parallel.

$$\boxed{S = \frac{I_g G}{I - I_g}}. \tag{64}$$



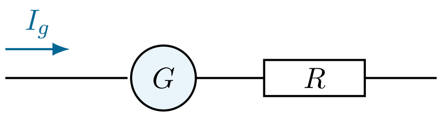
Low shunt carries most of the current.

Figure 14: Galvanometer converted into ammeter by parallel shunt resistance.

14.5 Conversion to Voltmeter

A galvanometer is converted to a voltmeter of range V by connecting a high resistance R in series.

$$\boxed{R = \frac{V}{I_g} - G}. \tag{65}$$



High series resistance limits current through the galvanometer.

Figure 15: Galvanometer converted into voltmeter by high series resistance.

15 Important Formula Sheet

Formula Box

$$\begin{aligned}
\boldsymbol{F}_B &= q(\boldsymbol{v} \times \boldsymbol{B}) \\
F_B &= |q|vB\sin\theta \\
\boldsymbol{F} &= q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B}) \\
r &= \frac{mv}{|q|B} \\
\omega &= \frac{|q|B}{m} \\
T &= \frac{2\pi m}{|q|B} \\
f &= \frac{|q|B}{2\pi m} \\
p &= v_{\parallel} T \\
\mathrm{d}\boldsymbol{B} &= \frac{\mu_0}{4\pi}\frac{I\,\mathrm{d}\boldsymbol{l} \times \hat{\boldsymbol{r}}}{r^2} \\
B_{\text{loop axis}} &= \frac{\mu_0 I R^2}{2(R^2 + x^2)^{3/2}} \\
B_{\text{loop centre}} &= \frac{\mu_0 I}{2R} \\
\oint \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{l} &= \mu_0 I_{\text{enc}}
\end{aligned}$$

$$\begin{aligned}
B_{\text{wire}} &= \frac{\mu_0 I}{2\pi r} \\
B_{\text{solenoid}} &\approx \mu_0 n I \\
\boldsymbol{F} &= I(\boldsymbol{L} \times \boldsymbol{B}) \\
F &= ILB\sin\theta \\
\frac{F}{L} &= \frac{\mu_0 I_1 I_2}{2\pi d} \\
\boldsymbol{m} &= NI\boldsymbol{A} \\
\boldsymbol{\tau} &= \boldsymbol{m} \times \boldsymbol{B} \\
\tau &= NIAB\sin\theta \\
I &= \frac{k}{NAB}\phi \\
S_I &= \frac{NAB}{k} \\
S &= \frac{I_g G}{I - I_g} \\
R &= \frac{V}{I_g} - G
\end{aligned}$$

16 Important Derivations for Board Exams

Derivation	Must include	Marks
Magnetic force circular motion	$qvB = mv^2/r$, r, T, f	3
Cyclotron frequency	Circular motion in dee, $f = \lvert q\rvert B/(2\pi m)$	3
Circular loop axial field	Biot-Savart, symmetry cancellation, axial component	5
Ampere's law for straight wire	Circular Amperian loop, $B(2\pi r) = \mu_0 I$	3
Force between parallel wires	Field due to wire 1, force on wire 2, F/L	3
Torque on current loop	Force pair, couple arm, $\tau = NIAB\sin\theta$	3/5
Moving coil galvanometer	Deflecting torque, restoring torque, $I \propto \phi$	3
Ammeter and voltmeter conversion	Current division or series resistance relation	2/3

17 Solved Examples

Example 1: Force on a Moving Charge

A proton moves with speed $2.0 \times 10^6\,\mathrm{m\,s^{-1}}$ perpendicular to a magnetic field of $0.50\,\mathrm{T}$. Find the magnetic force.
Solution.

$$\begin{aligned} F &= qvB\sin 90^\circ \\ &= (1.6 \times 10^{-19})(2.0 \times 10^6)(0.50) \\ &= 1.6 \times 10^{-13}\,\mathrm{N}. \end{aligned}$$

Answer: $1.6 \times 10^{-13}\,\mathrm{N}$.

Example 2: Radius of Circular Path

An electron of speed $3.0 \times 10^6\,\mathrm{m\,s^{-1}}$ enters a uniform magnetic field $2.0 \times 10^{-3}\,\mathrm{T}$ perpendicular to it. Find the radius of the path. Take $m_e = 9.1 \times 10^{-31}\,\mathrm{kg}$.
Solution.

$$\begin{aligned} r &= \frac{mv}{|q|B} \\ &= \frac{(9.1 \times 10^{-31})(3.0 \times 10^6)}{(1.6 \times 10^{-19})(2.0 \times 10^{-3})} \\ &= 8.53 \times 10^{-3}\,\mathrm{m}. \end{aligned}$$

Answer: $8.5\,\mathrm{mm}$ approximately.

Example 3: Field Due to Long Straight Wire

A long straight wire carries $5\,\mathrm{A}$. Find B at a point $2\,\mathrm{cm}$ from the wire.

Solution.

$$\begin{aligned} B &= \frac{\mu_0 I}{2\pi r} \\ &= \frac{4\pi \times 10^{-7} \times 5}{2\pi \times 0.02} \\ &= 5.0 \times 10^{-5}\,\mathrm{T}. \end{aligned}$$

Example 4: Force Between Parallel Wires

Two long parallel wires carry 10 A and 20 A in the same direction. They are separated by 5 cm. Find force per metre.
Solution.

$$\begin{aligned} \frac{F}{L} &= \frac{\mu_0 I_1 I_2}{2\pi d} \\ &= \frac{4\pi \times 10^{-7} \times 10 \times 20}{2\pi \times 0.05} \\ &= 8.0 \times 10^{-4}\,\mathrm{N\,m^{-1}}. \end{aligned}$$

Same direction currents attract.

Example 5: Galvanometer Conversion

A galvanometer has resistance $G = 30\,\Omega$ and full-scale current $I_g = 2\,\mathrm{mA}$. Find shunt resistance needed to convert it into an ammeter of range 1 A.
Solution.

$$\begin{aligned} S &= \frac{I_g G}{I - I_g} \\ &= \frac{(2 \times 10^{-3})(30)}{1 - 2 \times 10^{-3}} \\ &= 0.0601\,\Omega. \end{aligned}$$

Answer: $S \approx 0.060\,\Omega$.

18 Important Numericals for Practice

1. A charge $5\,\mu\mathrm{C}$ moves with speed $4 \times 10^5\,\mathrm{m\,s^{-1}}$ at $30°$ to a magnetic field 0.2 T. Find magnetic force.
2. An alpha particle enters a 0.1 T magnetic field perpendicular to it with speed $2 \times 10^6\,\mathrm{m\,s^{-1}}$. Find radius. Use $m_\alpha = 6.64 \times 10^{-27}\,\mathrm{kg}$ and $q_\alpha = 3.2 \times 10^{-19}\,\mathrm{C}$.
3. A circular coil of radius 10 cm has 50 turns and carries 2 A. Find magnetic field at its centre.
4. A wire carrying 8 A produces field at 4 cm. Calculate B.
5. Two parallel conductors carry 4 A and 6 A in opposite directions. Distance between them is 3 cm. Find force per metre and state whether it is attractive or repulsive.
6. A rectangular coil of 100 turns and area $2 \times 10^{-3}\,\mathrm{m^2}$ carries current 0.5 A in magnetic field 0.4 T. Find maximum torque.
7. A galvanometer has $G = 50\,\Omega$ and $I_g = 1\,\mathrm{mA}$. Find the series resistance to convert it into a 5 V voltmeter.

Answer Key

1. 0.20 N
2. 0.415 m
3. 6.28×10^{-4} T
4. 4.0×10^{-5} T
5. 1.6×10^{-4} N m^{-1}, repulsive
6. 4.0×10^{-2} N m
7. 4950 Ω

19 NCERT-Based Questions

Conceptual Questions

1. Why does a magnetic field not change the speed of a charged particle?
2. A charged particle moves parallel to a uniform magnetic field. What is its path? Justify.
3. What is the direction of magnetic field at the centre of a circular loop carrying clockwise current as seen by an observer?
4. Why are pole pieces in a moving coil galvanometer made concave?
5. Why is shunt resistance connected in parallel for ammeter conversion?
6. Why is high resistance connected in series for voltmeter conversion?
7. State the condition under which a charged particle remains undeflected in crossed electric and magnetic fields.

NCERT-Style Numerical Questions

1. Calculate the magnetic field at the centre of a 20 turn circular coil of radius 0.1 m carrying 0.5 A.
2. A proton moves in a circular path of radius 0.5 m in a magnetic field 0.2 T. Find its speed.
3. A straight wire carries 10 A. At what distance from it is magnetic field 2×10^{-5} T?
4. A galvanometer with $G = 25\,\Omega$ and $I_g = 4$ mA is to be converted into an ammeter of range 2 A. Find shunt resistance.

20 Previous-Year Style Board Questions

1-Mark Questions

1. Write the SI unit of magnetic field.
2. State the condition for maximum magnetic force on a moving charge.
3. What is the magnetic field at the centre of a circular loop of radius R carrying current I?
4. Define current sensitivity of a moving coil galvanometer.
5. In what direction do two parallel wires carrying currents in opposite directions exert force on each other?

2-Mark Questions

1. State Biot-Savart law. Mention the direction of magnetic field due to a current element.
2. Derive the expression for the radius of a charged particle moving perpendicular to a uniform magnetic field.
3. Draw a labeled diagram of a moving coil galvanometer and state its principle.

4. Explain why a charged particle moving in a magnetic field has constant kinetic energy.

3-Mark Questions

1. Using Ampere's circuital law, derive magnetic field due to an infinitely long straight current-carrying wire.
2. Derive the expression for force per unit length between two long parallel current-carrying conductors.
3. Derive cyclotron frequency and explain why it is independent of speed.
4. Explain conversion of galvanometer into ammeter and derive the shunt resistance formula.

5-Mark Questions

1. Derive magnetic field on the axis of a circular current loop using Biot-Savart law. Write the field at the centre.
2. Derive torque on a rectangular current loop in a uniform magnetic field. Define magnetic dipole moment of the loop.
3. Explain construction, principle, and working of a moving coil galvanometer. Derive relation between current and deflection.

21 Common Mistakes and Corrections

Common Mistake

Mistake: Using qvB without $\sin\theta$.
Correction: Always write $F = |q|vB\sin\theta$. Use qvB only when $\theta = 90°$.

Common Mistake

Mistake: Forgetting to reverse force direction for an electron.
Correction: Right-hand rule gives direction for positive charge. For negative charge, reverse it.

Common Mistake

Mistake: Confusing radius of circular loop R with distance from wire r.
Correction: Use R for loop radius and r or d for distance from straight wire.

Common Mistake

Mistake: Saying magnetic force changes kinetic energy.
Correction: Magnetic force is perpendicular to velocity, so work done is zero and kinetic energy is unchanged.

Common Mistake

Mistake: Connecting ammeter conversion resistance in series.
Correction: Ammeter needs low resistance; connect shunt in parallel. Voltmeter needs high resistance; connect series resistance.

22 Quick Revision Notes

- Magnetic force on charge: $\boldsymbol{F} = q\boldsymbol{v} \times \boldsymbol{B}$.
- Magnetic force is zero if charge is stationary or if $\boldsymbol{v} \parallel \boldsymbol{B}$.

- Uniform magnetic field perpendicular to velocity produces circular motion.
- Radius $r = mv/(|q|B)$, time period $T = 2\pi m/(|q|B)$.
- Cyclotron frequency: $f = |q|B/(2\pi m)$.
- Biot-Savart law gives magnetic field due to current element.
- Circular loop centre field: $B = \mu_0 NI/(2R)$.
- Ampere's law: $\oint \boldsymbol{B} \cdot d\boldsymbol{l} = \mu_0 I_{\text{enc}}$.
- Infinite straight wire: $B = \mu_0 I/(2\pi r)$.
- Same direction currents attract; opposite direction currents repel.
- Torque on coil: $\tau = NIAB \sin\theta$.
- Moving coil galvanometer: $I = (k/NAB)\phi$.
- Ammeter: low shunt in parallel. Voltmeter: high resistance in series.

23 Final Revision Checklist

No.	Task	Done
1	I can apply right-hand rule and dot-cross convention correctly.	☐
2	I can derive r, T, and f for circular motion in a magnetic field.	☐
3	I can explain Lorentz force and velocity selector condition.	☐
4	I can draw and explain cyclotron with frequency derivation.	☐
5	I can state Biot-Savart law in vector and scalar form.	☐
6	I can derive field on the axis of a circular current loop.	☐
7	I can use Ampere's law to derive field due to long straight wire.	☐
8	I can solve numericals on $B = \mu_0 I/(2\pi r)$ and loop field.	☐
9	I can derive force per unit length between parallel currents.	☐
10	I can derive torque on current loop and define magnetic moment.	☐
11	I can explain moving coil galvanometer and current sensitivity.	☐
12	I can derive shunt/series resistance for ammeter/voltmeter conversion.	☐

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One-Page Last-Minute Formula Recap

Concept	Formula
Magnetic force on charge	$\boldsymbol{F} = q(\boldsymbol{v} \times \boldsymbol{B}),\ F = \lvert q\rvert vB\sin\theta$
Lorentz force	$\boldsymbol{F} = q(\boldsymbol{E} + \boldsymbol{v} \times \boldsymbol{B})$
Circular motion	$r = mv/(\lvert q\rvert B),\ T = 2\pi m/(\lvert q\rvert B),\ f = \lvert q\rvert B/(2\pi m)$
Biot-Savart law	$\mathrm{d}\boldsymbol{B} = (\mu_0/4\pi)(I\,\mathrm{d}\boldsymbol{l} \times \hat{\boldsymbol{r}})/r^2$
Loop axis field	$B = \mu_0 NIR^2/[2(R^2 + x^2)^{3/2}]$
Loop centre	$B = \mu_0 NI/(2R)$
Ampere's law	$\oint \boldsymbol{B} \cdot \mathrm{d}\boldsymbol{l} = \mu_0 I_{\mathrm{enc}}$
Straight wire	$B = \mu_0 I/(2\pi r)$
Current conductor force	$\boldsymbol{F} = I(\boldsymbol{L} \times \boldsymbol{B})$
Parallel wires	$F/L = \mu_0 I_1 I_2/(2\pi d)$
Torque on loop	$\boldsymbol{\tau} = \boldsymbol{m} \times \boldsymbol{B},\ \tau = NIAB\sin\theta$
MCG current relation	$I = k\phi/(NAB)$
Ammeter shunt	$S = I_g G/(I - I_g)$
Voltmeter conversion resistance	$R = V/I_g - G$

End of Chapter 4 Notes - Revise formulas, diagrams, and derivations before attempting board-style papers.